

Keywords of Plotfunction

Miguel Valbuena

April 26, 2026

Contents

1	Colors	7
1.0.1	beige	7
1.0.2	black	7
1.0.3	blue	7
1.0.4	brown	7
1.0.5	cyan	7
1.0.6	dblue	7
1.0.7	dgreen	7
1.0.8	gold	7
1.0.9	green	7
1.0.10	grey	7
1.0.11	lblue	7
1.0.12	lgreen	7
1.0.13	lgrey	7
1.0.14	magenta	8
1.0.15	olive	8
1.0.16	orange	8
1.0.17	pink	8
1.0.18	purple	8
1.0.19	red	8
1.0.20	skyblue	8
1.0.21	violet	8
1.0.22	white	8
1.0.23	yellow	8

2	Commands	8
2.1	analyze:	8
2.2	arclength:	8
2.3	calculate:	8
2.4	derive:	9
2.5	evaluate:	9
2.6	integral:	9
2.7	interpolation:	9
2.8	intersection:	9
2.9	latex-string:	9
2.10	move:	10
2.11	normal:	10
2.12	osc-circle:	10
2.13	points-in-graph:	10
2.14	reflection:	10
2.15	regression:	10
2.16	reset-colors:	10
2.17	rotate:	11
2.18	show-function:	11
2.19	show-grid:	11
2.20	show-x-axis:	11
2.21	show-x-scale:	11
2.22	show-y-axis:	11
2.23	show-y-scale:	11
2.24	table:	11
2.25	tangent:	12
2.26	taylor-polynomial:	12
2.27	zero-iteration:	12
3	Constants	12
3.0.1	inf:	12
3.0.2	PI:	12
4	Geometric Objects	12
4.1	bitmap:	12
4.2	circle:	12
4.3	fill:	13
4.4	histogram:	13
4.5	line	13
4.6	points:	13

4.7	polygon:	13
4.8	rectangle:	13
4.9	text:	14
4.10	triangle:	14
5	Functions	14
5.1	abs:	14
5.2	arccos:	14
5.3	arcosh:	14
5.4	arcsin:	14
5.5	arctan:	14
5.6	arsinh:	14
5.7	artanh:	14
5.8	cos:	15
5.9	cosh:	15
5.10	exp:	15
5.11	log:	15
5.12	sin:	15
5.13	sinh:	15
5.14	sqrt:	15
5.15	tan:	15
5.16	tanh:	15
6	Settings	15
6.1	axis-color:	15
6.2	axis-fontsize:	16
6.3	axis-names:	16
6.4	axis-range:	16
6.5	axis-scale:	16
6.6	background-color:	16
6.7	font-size:	16
6.8	function-color:	16
6.9	function-settings:	16
6.10	function-size:	17
6.11	grid-color:	17
6.12	grid-scale:	17
6.13	margin:	17
6.14	move-increment:	17
6.15	panel-size:	17
6.16	parameter-values:	17

6.17	point-decimals:	17
6.18	point-size:	18
6.19	regression-degree:	18
6.20	select:	18
6.21	values-number:	18
7	Alphabetically	20
7.1	A	20
7.1.1	abs	20
7.1.2	analyze:	20
7.1.3	arccos	20
7.1.4	arclength	20
7.1.5	arcosh	20
7.1.6	arcsin	20
7.1.7	arctan	20
7.1.8	arsinh	20
7.1.9	artanh	20
7.1.10	axis-color	20
7.1.11	axis-fontsize	20
7.1.12	axis-names	20
7.1.13	axis-range	20
7.1.14	axis-scale	20
7.2	B	20
7.2.1	background-color	20
7.2.2	beige	20
7.2.3	bitmap	20
7.2.4	black	20
7.2.5	blue	20
7.2.6	brown	20
7.3	C	20
7.3.1	calculate	20
7.3.2	cos	20
7.3.3	cosh	20
7.3.4	cyan	20
7.4	D	20
7.4.1	dblue	20
7.4.2	derive	20
7.4.3	dgreen	20
7.5	E	20
7.5.1	evaluate	20

	7.5.2	exp	20
7.6	F		20
	7.6.1	fill	20
	7.6.2	font-size	20
	7.6.3	function-color	20
	7.6.4	function-settings	20
	7.6.5	function-size	20
7.7	G		20
	7.7.1	gold	20
	7.7.2	green	20
	7.7.3	grey	20
	7.7.4	grid-color	20
	7.7.5	grid-scale	20
7.8	H		20
	7.8.1	histogram	20
7.9	I		20
	7.9.1	inf	20
	7.9.2	integral	20
	7.9.3	interpolation	20
	7.9.4	intersection	20
7.10	L		20
	7.10.1	latex-string	20
	7.10.2	lblue	20
	7.10.3	lgreen	20
	7.10.4	lgrey	20
	7.10.5	line	20
	7.10.6	log	20
7.11	M		20
	7.11.1	magenta	20
	7.11.2	margin	20
	7.11.3	move	20
	7.11.4	move-increment	20
7.12	N		20
	7.12.1	normal	20
7.13	O		20
	7.13.1	olive	20
	7.13.2	orange	20
	7.13.3	osc-circle	20
7.14	P		20
	7.14.1	panel-size	20

7.14.2	parameter-values	20
7.14.3	pink	20
7.14.4	PI	20
7.14.5	point-decimals	20
7.14.6	point-size	20
7.14.7	points	20
7.14.8	points-in-graph	20
7.14.9	polygon	20
7.14.10	purple	20
7.15	R	20
7.15.1	rectangle	20
7.15.2	red	20
7.15.3	reflection	20
7.15.4	regression	20
7.15.5	regression-degree	20
7.15.6	reset-colors	20
7.15.7	rotate	20
7.16	S	20
7.16.1	select	20
7.16.2	show-function	20
7.16.3	show-grid	20
7.16.4	show-x-axis	20
7.16.5	show-x-scale	20
7.16.6	show-y-axis	20
7.16.7	show-y-scale	20
7.16.8	sin	20
7.16.9	sinh	20
7.16.10	skyblue	20
7.16.11	sqrt	20
7.17	T	20
7.17.1	table	20
7.17.2	tan	20
7.17.3	tangent	20
7.17.4	tanh	20
7.17.5	taylor-polynomial	20
7.17.6	text	20
7.17.7	triangle	20
7.18	V	20
7.18.1	values-number	20
7.18.2	violet	20

7.19	W	20
7.19.1	white	20
7.20	Y	20
7.20.1	yellow	20
7.21	Z	20
7.21.1	zero-iteration	20

1 Colors

1.0.1 beige

1.0.2 black

1.0.3 blue

1.0.4 brown

1.0.5 cyan

1.0.6 dblue

Dark blue.

1.0.7 dgreen

Dark green

1.0.8 gold

1.0.9 green

1.0.10 grey

1.0.11 lblue

light blue

1.0.12 lgreen

light green

1.0.13 lgrey

light grey

1.0.14 magenta

1.0.15 olive

1.0.16 orange

1.0.17 pink

1.0.18 purple

1.0.19 red

1.0.20 skyblue

1.0.21 violet

1.0.22 white

1.0.23 yellow

2 Commands

2.1 analyze:

Usage: **analyze** [#nr = 1] [x₁ x₂] [prec = 1e-9] [iterations = number of pixels of width][decimals = 4]

Analyses the function number nr in the [x₁ , x₂] interval. prec is the precision used in the algorithms. Iteration is the number of evaluations of that function to count the changes of sign in the given interval. decimals are the number of decimals for rounding all results.

2.2 arclength:

Usage: **arclength** [#nr = 1 / “function”] [decimals=4 iterations=50 precision=1e-8] [x₁ x₂]

Calculates the arc length of the function #nr or “function” in the interval [x₁ x₂] if given, or the x axis interval (default). The approximate result is rounded by the given decimals. Iterations and precision are needed in the used algorithms to approximate the integral value (Kepler Simpson).

2.3 calculate:

Usage: **calculate** arithmetic expression

2.4 derive:

Usage **derive** [#nr = 1]

Derives function to x (x_1).

2.5 evaluate:

Usage **evaluate** [#nr = 1] expressions [decimals = 4]

Evaluates the #nr-th. function value of the expressions and rounds the result to the given decimals.

2.6 integral:

Usage: **integral** [#nr = 1 / “term”] [decimals iterations precision] [x_1 x_2]

Numerical approximation (if possible also exact computation via a stamm-function) of the integral of the #nr-th function (or the function by the given term) in the interval [x_1 x_2]. The result is rounded to the given decimals and the algorithms use the given number or iterations and precision (Kepler-Simpson-Algorithm).

2.7 interpolation:

Usage: **interpolation** #nr / points for f [; points for f'] [;points for f'']

Polynomial interpolation of the #nr-th geometric object (points or polygon) or the given points for f and eventually f' and f''.

2.8 intersection:

Usage: **intersection** [#nr1 = 1] #nr2 [x_1 x_2] [precision] [iterations] [decimals]

Computes the approximation (if possible also the exact) points of intersection and intersection angle between the #nr1-th and #nr2-th object in the [x_1 x_2] interval. The coordinates are rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

2.9 latex-string:

Usage: **latex-string** [#nr = 1]

Translates the #nr-th function term to a latex string. Example: *latex-string* $x/exp(x)$ would compute the string: $\frac{x}{e^x}$

2.10 move:

Usage: **move** [#nr = 1] dx dy

Moves the #nr-th geometric object dx units to the right (left if $dx < 0$) and dy units to the top (bottom if $dy < 0$).

2.11 normal:

Usage: **normal** [#nr = 1] x / x y

Computes the normal to the #nr-th function at x or from the point (x,y).

2.12 osc-circle:

Usage: **osc-circle** [#nr = 1] x

Computes the osculating circle to the #nr-th function (or parametric curve) at x.

2.13 points-in-graph:

Usage: **points-in-graph** [#nr = 1] expression1 (x= / y=); expression2... [;ddecimal [= 4]]

Draws the points in the graph of the nr-th function wrt the given x-, y- or parameter-expressions.

Example: *points-in-graph x=4; y=sqrt(2); d=2* draws the points to the graph of the first function wrt to $x = 4$ and $y = \sqrt{2}$. The number of decimals for these points is 2.

2.14 reflection:

Usage: **reflection** [#nr1 = 1] #nr2 / object

Reflects the object nr1 to the axis/point defined by the object nr2.

2.15 regression:

Usage: **regression** #nr [= 1]/points [degree = regression-degree]

Computes the regression polynomial of the given degree wrt the given points or the #nr-th geometric object.

2.16 reset-colors:

Resets all the colors to the default colors.

2.17 rotate:

Usage: **rotate** [#nr = 1] angle (in degrees) x y (center)

Rotates the #nr-th geometric object / function by an angle of the given degrees wrt the given center (x,y).

2.18 show-function:

Usage: **show-function** [#nr = 1] 0/1

Shows (if value is 1) or hides (if value is 0) the #nr-th object.

2.19 show-grid:

Usage: **show-grid** 0/1

Shows (if value is 1) or hides (if value is 0) the grid.

2.20 show-x-axis:

Usage: **show-x-axis** 0/1

Shows (if value is 1) or hides (if value is 0) the x-axis.

2.21 show-x-scale:

Usage: **show-x-scale** 0/1

Shows (if value is 1) or hides (if value is 0) the x-scale.

2.22 show-y-axis:

Usage: **show-y-axis** 0/1

Shows (if value is 1) or hides (if value is 0) the y-axis.

2.23 show-y-scale:

Usage: **show-y-scale** 0/1

Shows (if value is 1) or hides (if value is 0) the y-scale.

2.24 table:

Usage: **table** [#nr = 1 / “function”] [x₁ x₂] [dx = 0.5] [;]

Evaluates the function in the values from x₁ to x₂ with an increment of dx.

If there is a ; at the end, the values are printed in fractions.

2.25 tangent:

Usage: **tangent** [#nr = 1] x / x y

Computes the tangent to the #nr-th function at x or from the point (x,y).

2.26 taylor-polynomial:

Usage: **taylor-polynomial** [#nr = 1] deg [x0 = 0]

Computes the taylor polynomial or deg-th degree of the nr-th function at x0.

2.27 zero-iteration:

Usage: *zero-iteration [#nr = 1 / “function”] [decimals iterations precision]
[x₁ x₂] *

Computes the approximation of a root in the [x₁, x₂] interval of the given object.

The result is rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

3 Constants

3.0.1 inf:

stands for infinity.

3.0.2 PI:

stands for the irrational number π .

4 Geometric Objects

4.1 bitmap:

Usage: **bitmap_**[nr] x y width height

Draws the stored bitmap nr on to the position x y (axis-coordinates) with the given width and height in pixels.

4.2 circle:

Usage: **circle** x y r [α_1 α_2 [slice]] [<color>]

Draws a circle with middle-point x y, radius r. If α_1 , α_2 is given an arc of

circle is drawn, beginning at the angle α_1 and ending at α_2 . If slice = 1 a transparent slice is drawn, if slice = 2 a filled slice is drawn.

4.3 fill:

Usage: **fill** **x** **y** [**<color>**]

Floodfill at the point (x,y) with the given color.

4.4 histogram:

Usage: **histogram** **points** [**t = 0** , **w = 1**] [**<color>**]

Draws a histogram wrt the given points. t is the value for the transparency of the background color, where 0 is transparent and 1 is opaque. w is the width of the columns wrt the x-axis range.

4.5 line

Usage: **line** **point₁** **point₂** [**h = 0**] [**<color,>** **l**]

Draws a line (or arrow if h != 0) from point₁ to point₂. h is a float number for the size of the arrowhead (h = 0 for no arrowhead). If l >= 1 the line will be dotted.

4.6 points:

Usage: **points** **x₁** **y₁** **x₂** **y₂** ... **x_n** **y_n** [**<color>**]

Draws n points with the given coordinates.

4.7 polygon:

Usage: **polygon** **x₁** **y₁** **x₂** **y₂** ... **x_n** **y_n** [**<color>**]

Draws polygon with the n vertices (x₁,y₁), ... , (x_n,y_n).

4.8 rectangle:

Usage: **rectangle** **x₁** **y₁** **x₂** **y₂**

Draws a rectangle with the left bottom corner (x₁,y₁) and the top right corner (x₂,y₂).

4.9 text:

Usage: **text** {[l] string} x y [<color>]

Displays a string at the coordinates (x,y).

If latex is installed you can use `\to` generate a latex string.

4.10 triangle:

Usage: **triangle** x₁ y₁ x₂ y₂ x₃ y₃ [<[color,] l>]

Draws a triangle with vertices (x₁,y₁), (x₂,y₂), (x₃,y₃).

If l >= 1 the line will be dotted.

5 Functions

5.1 abs:

Absolute value of a real or a complex number.

5.2 arccos:

Arcus cosine function.

5.3 arcosh:

Area hyperbolic cosine function.

5.4 arcsin:

Arcus sine function.

5.5 arctan:

Arcus tangent function.

5.6 arsinh:

Area hyperbolic sine function.

5.7 artanh:

Area hyperbolic tangent function.

5.8 cos:

5.9 cosh:

Hyperbolic cosine.

5.10 exp:

The exponential function.

5.11 log:

Logarithmus Naturalis.

5.12 sin:

Sine function.

5.13 sinh:

Hyperbolic cosine.

5.14 sqrt:

The square root function.

5.15 tan:

The trigonometric tangent function.

5.16 tanh:

Hyperbolic tangent function.

6 Settings

6.1 axis-color:

Usage: **def. color / Red Green Blue**

Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

6.2 axis-fontsize:

Usage **axis-fontsize** unsigned int

6.3 axis-names:

Usage: **axis-names** string string

Example: *axis-names x y*

6.4 axis-range:

Usage: **axis-range** x_1 x_2 [: y_1 y_2]

Example: *axis-range -3 5 : -2 7* : x values range from -3 to 5, y values from -2 to 7
axis-range 5 : 10 : x values from -5 to 5, y values from -10 to 10.

6.5 axis-scale:

Usage : **axis-scale** s_x [$s_y = s_x$]

Example: *axis-scale $PI/2$ 1*

6.6 background-color:

Usage: **background-color** def. color / Red Green Blue

Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

6.7 font-size:

Usage: **font-size** unsigned int

Sets the font size in pixels for input/output text.

6.8 function-color:

Usage: **function-color** [#nr = 1] defined color / Red Green Blue

Draws the #nr-th function/geometric object with the given defined color or the color defined by the three values from 0 to 255 for the red, green and blue part.

6.9 function-settings:

Usage: **function-settings** [#nr = 1]

Opens a settings dialog for the #nr-th. function or geometric object.

6.10 **function-size:**

Usage: **function-size** [#nr = 1] **unsigned number**

Sets the line thickness of the #nr-th. function/geometric object to the given number.

6.11 **grid-color:**

Usage: **grid.-color** **def. color** / **Red Green Blue**

Draws the grid lines in the given defined or the color obtained by the three values for the red, green and blue part (from 0 to 255 each).

6.12 **grid-scale:**

Usage: **grid-scale** **sx** [**sy = sx**]

Example: *grid-scale* $PI/2$ 1

6.13 **margin:**

Usage: **margin** **unsigned int**

Sets the margin (offset) of the coordinates system in pixels.

6.14 **move-increment:**

Usage: **move-increment** **dx** [**dy=dx**] (**p** for **pixels**)

Move increment values for moving objects or the coordinate system with the arrow-keys.

6.15 **panel-size:**

Usage: **panel-size** **sx** [**sy = sx**]

6.16 **parameter-values:**

Usage: **parameter-values** **double ...**

Sets the values for the parameter t of the functions.

6.17 **point-decimals:**

Usage: **point-decimals** **int**

6.18 point-size:

Usage: **point-size int**

Sets the size in pixel for points (wrt to style 1 and 2)

6.19 regression-degree:

Usage: **regression-degree unsigned int**

Sets the regression degree for the regression computations.

6.20 select:

Usage: **select [#nr=1]**

Selects (highlights) the nr-th function.

6.21 values-number:

Usage **values-number unsigned int**

The number of points that are evaluated for rendering the functions.

7 Alphabetically

7.1 A

7.1.1 abs

7.1.2 analyze:

7.1.3 arccos

7.1.4 arclength

7.1.5 arcosh

7.1.6 arcsin

7.1.7 arctan

7.1.8 arsinh

7.1.9 artanh

7.1.10 axis-color

7.1.11 axis-fontsize

7.1.12 axis-names

7.1.13 axis-range

7.1.14 axis-scale

7.2 B

7.2.1 background-color

7.2.2 beige

7.2.3 bitmap

7.2.4 black

7.2.5 blue

7.2.6 brown

7.3 C

7.3.1 calculate

7.3.2 cos

7.3.3 cosh

7.3.4 cyan

7.4 D

7.4.1 dblue

7.4.2 derive

7.4.3 dgreen